JEREMIAH NYAGA NATIONAL POLYTECHNIC.

DEPARTMENT: ELECTRICAL AND ELECTRONIC ENGINEERING.

CLASS: CBET/EI/24J/6

END OF TERM EXAM

SUBJECT: ELECTRICAL PRINCIPLES

TIME: 2 HOURS.

INSTRUCTIONS:

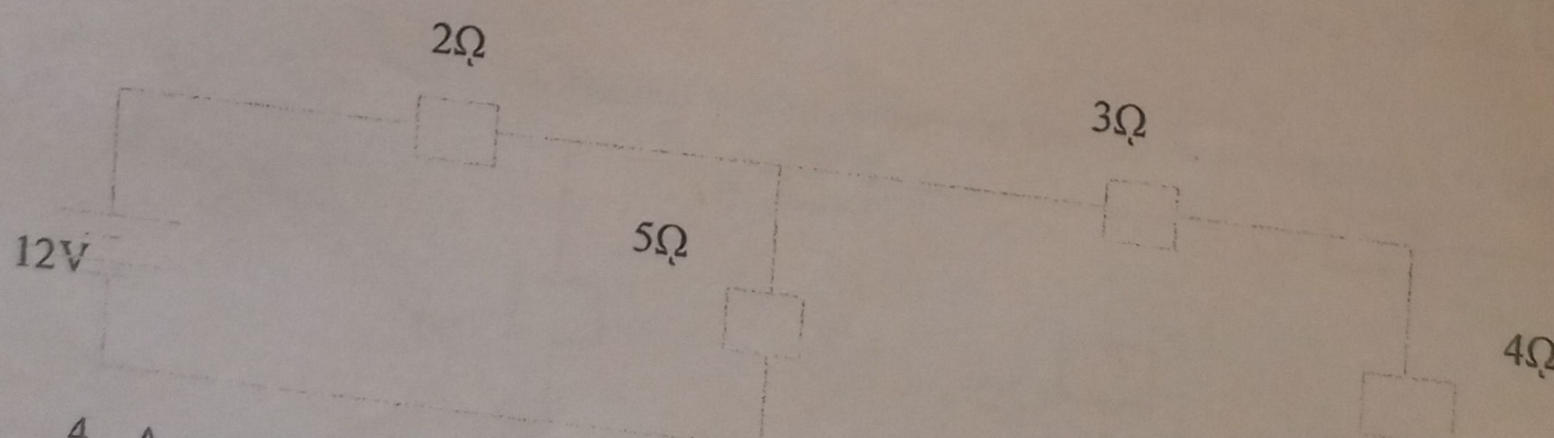
This test is out of 100 marks. Provide answers in the booklet provided.

Attempt <u>ALL</u> questions.

SECTION A (40 MARKS)

Answer all questions in this section.

1. Give the SI unit and symbol of the SI unit for the following electrical quantities.
   a. Length
   b. Impendence
   c. Inductance
   d. Time
   (4 marks)
2. State ohm's law. (2 marks)
3. The figure below shows resistors arrangement in a circuit. Calculate; (6 Marks)
   i. Effective resistance
   ii. Current through 5Ω resistor



4. A source of 5V supplies a current of 3A for 10 minutes. How much energy is provided in this time? (3 marks)
5. Giving two examples, define derived unit. (3 marks)
6. State three factors that affect resistance of a conductor. (3 marks)
7. Using electrical symbols differentiate between. (4 marks)
   i. A.C and D.C
   ii. Cell and Battery

8. With aid of a diagram, explain how an ammeter and voltmeter are connected in electric circuit. (4 marks)

9. Two resistors are connected in series across a 24V supply and a current of 6A flows in the circuit. If one of the resistors has a resistance of 2 ohms, determine; (5 marks)

a) The value of the other resistor

b) The P.D across the 2 ohms resistor

10. Table below shows electrical quantities and their S.I. unit. Fill out the table. (6 Marks)

Electrical Quantity	S.I Unit	Symbol for S.I Unit
Energy		J
	Amperes	
Charge		
	Siemens	S

SECTION B (60 MARKS)

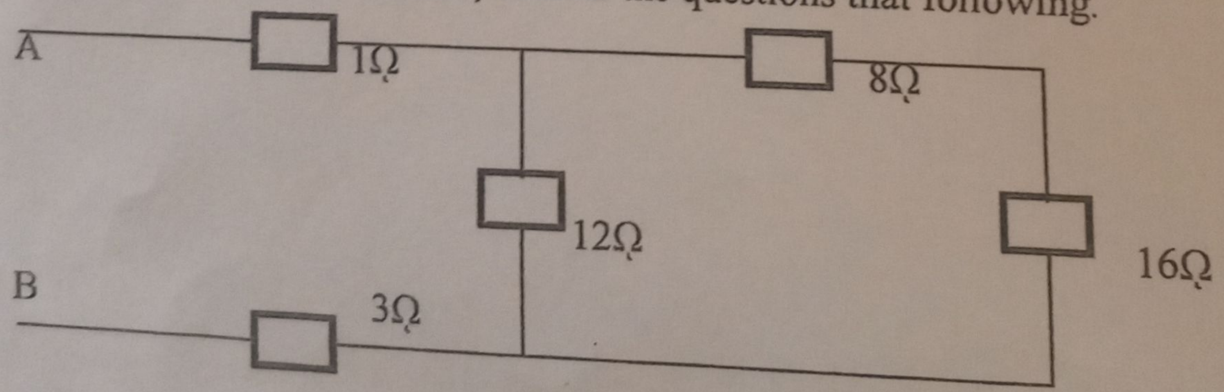
*Answer any **THREE** questions in this section.*

11. (a) State four components of an electrical circuit diagram. (4 Marks)

(b) Define the following laws; (4 Marks)

i. Kirchhoff's voltage law
ii. Kirchhoff's current law

(c) For the circuit below, answer the questions that following. (12 marks)



A 24V e.m.f is connected between terminals A and B. Calculate;

i) Total resistance

ii) Current through the circuit

iii) Current through 12Ω resistor

iv) Power consumed across 12Ω resistor

v) If the power in (iv) above acts for 2 minutes, what is the energy consumed. (4 Marks)

12. (a) Differentiate A.C conditioning from Refrigeration. (6 Marks)

(b) State six I.E.E regulations on earthing of systems. (10 Marks)

(b) Explain five factors to consider when selecting method of earthing. (10 Marks)

13. (a) Using diagrams, explain three methods/ types of earthing system. (10 Marks)

(b) Explain five components of a lightning protection system. (10 Marks)

14. (a) State five types of lightning arrestors. (5 Marks)

(b) State five importance of earthing a system. (5 Marks)

(c) Using a diagram, explain the components of an earthing system. (10 Marks)